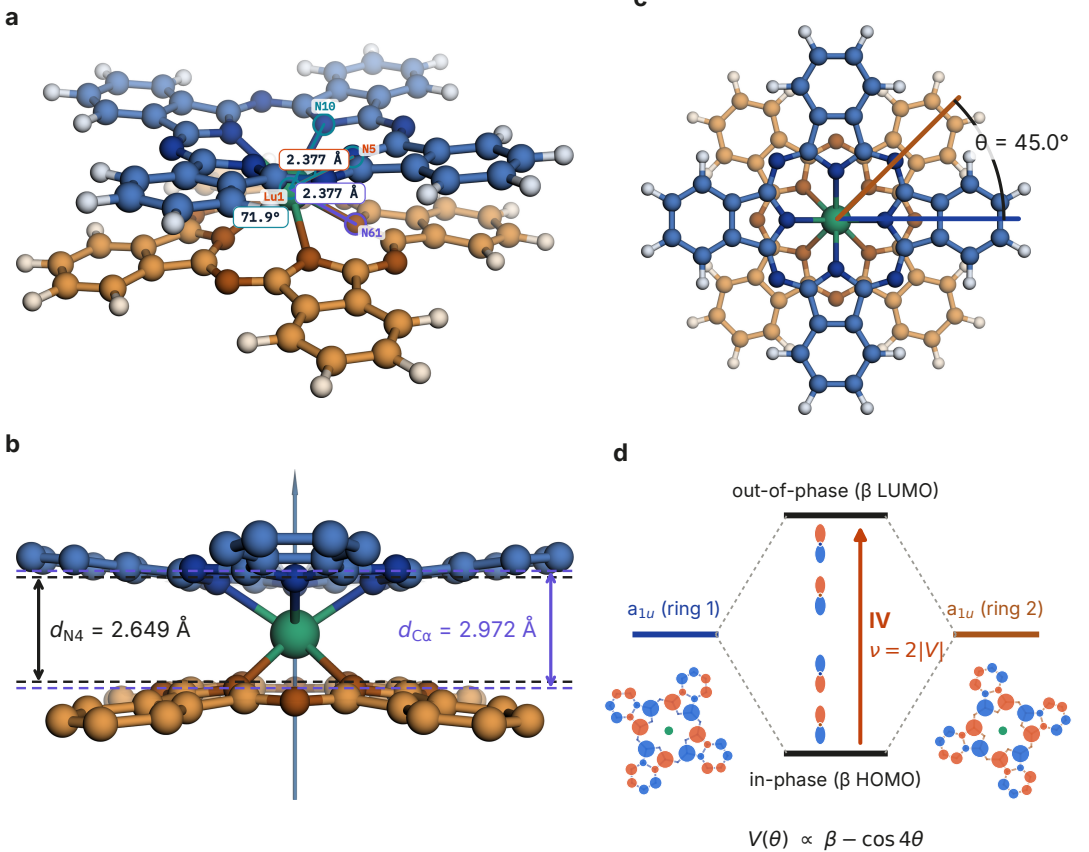

a
N10
N5
2.377 Å
Lu1
2.377 Å
71.9°
N61
b
d_{N4} = 2.649 Å
$d_{C\alpha}$ = 2.972 Å
c
θ = 45.0°
d
out-of-phase (β LUMO)
a_{1u} (ring 1)
IV
$\nu = 2|V|$
a_{1u} (ring 2)
in-phase (β HOMO)
$V(\theta) \propto \beta - \cos 4\theta$